

Exercise 4.2

Important Formulas

- (i) $(x + y)^2 = x^2 + 2xy + y^2$
- (ii) $(x - y)^2 = x^2 - 2xy + y^2$
- (iii) $x^2 - y^2 = (x + y)(x - y)$
- (iv) $(x + y)^3 = x^3 + y^3 + 3xy(x + y)$
- (v) $(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$
- (vi) $(x - y)^3 = x^3 - y^3 - 3xy(x - y)$
- (vii) $(x + y)^3 = x^3 - 3x^2y + 3xy^2 - y^3$
- (viii) $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$
- (ix) $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$

1. Factorize each of the following expressions:

(i) $4x^4 + 81y^4$

$$\begin{aligned}
 &4x^4 + 81y^4 \\
 &= (2x^2)^2 + (9y^2)^2 + 2(2x^2)(9y^2) - 36x^2y^2 \\
 &= (2x^2 + 9y^2)^2 - (6xy)^2 \\
 &= (2x^2 + 9y^2 + 6xy)(2x^2 + 9y^2 - 6xy) \\
 &= (2x^2 + 6xy + 9y^2)(2x^2 - 6xy + 9y^2)
 \end{aligned}$$

(ii) $a^4 + 64b^4$

$$\begin{aligned}
 &a^4 + 64b^4 \\
 &= (a^2)^2 + (8b^2)^2 + 2(a^2)(8b^2) - 16a^2b^2 \\
 &= (a^2 + 8b^2)^2 - (4ab)^2 \\
 &= (a^2 + 8b^2 + 4ab)(a^2 + 8b^2 - 4ab) \\
 &= (a^2 + 4ab + 8b^2)(a^2 - 4ab + 8b^2)
 \end{aligned}$$

(iii) $x^4 + 4x^2 + 16$

$$\begin{aligned}
 &x^4 + 4x^2 + 16 \\
 &= x^4 + 16 + 4x^2 \\
 &= (x^2)^2 + (4)^2 + 2(x^2)(4) - 8x^2 + 4x^2 \\
 &= (x^2 + 4)^2 - 4x^2 \\
 &= (x^2 + 4)^2 - (2x)^2 \\
 &= (x^2 + 4 + 2x)(x^2 + 4 - 2x) \\
 &= (x^2 + 2x + 4)(x^2 - 2x + 4)
 \end{aligned}$$

(iv) $x^4 - 14x^2 + 1$

$$\begin{aligned}
 &x^4 - 14x^2 + 1 \\
 &= x^4 + 1 - 14x^2 \\
 &= (x^2)^2 + (1)^2 + 2(x^2)(1) - 2x^2 - 14x^2 \\
 &= (x^2 + 1)^2 - 16x^2 \\
 &= (x^2 + 1)^2 - (4x)^2 \\
 &= (x^2 + 1 + 4x)(x^2 + 1 - 4x) \\
 &= (x^2 + 4x + 1)(x^2 - 4x + 1)
 \end{aligned}$$

(v) $x^4 - 30x^2y^2 + 9y^4$

$$\begin{aligned}
 &x^4 - 30x^2y^2 + 9y^4 \\
 &= x^4 + 9y^4 - 30x^2y^2 \\
 &= (x^2)^2 + (3y^2)^2 + 2(x^2)(3y^2) - 6x^2y^2 - 30x^2y^2 \\
 &= (x^2 + 3y^2)^2 - 36x^2y^2
 \end{aligned}$$

$$\begin{aligned}
 &= (x^2 + 3y^2)^2 - (6xy)^2 \\
 &= (x^2 + 3y^2 + 6xy)(x^2 + 3y^2 - 6xy) \\
 &= (x^2 + 6xy + 3y^2)(x^2 - 6xy + 3y^2)
 \end{aligned}$$

(vi) $x^4 - 7x^2y^2 + y^4$

$$\begin{aligned}
 &x^4 - 7x^2y^2 + y^4 \\
 &= x^4 + y^4 - 7x^2y^2 \\
 &= (x^2)^2 + (y^2)^2 + 2(x^2)(y^2) - 2x^2y^2 - 7x^2y^2 \\
 &= (x^2 + y^2)^2 - 9x^2y^2 \\
 &= (x^2 + y^2)^2 - (3xy)^2 \\
 &= (x^2 + y^2 + 3xy)(x^2 + y^2 - 3xy) \\
 &= (x^2 + 3xy + y^2)(x^2 - 3xy + y^2)
 \end{aligned}$$

2. Factorize each of the following expressions:

(i) $(x + 1)(x + 2)(x + 3)(x + 4) + 1$

$$(x + 1)(x + 2)(x + 3)(x + 4) + 1$$

$$1 + 4 = 5 = 2 + 3$$

$$\begin{aligned}
 &= [(x + 1)(x + 4)][(x + 2)(x + 3)] + 1 \\
 &= [x(x + 4) + 1(x + 4)][x(x + 3) + 2(x + 3)] + 1 \\
 &= [x^2 + 4x + x + 4][x^2 + 3x + 2x + 6] + 1 \\
 &= [x^2 + 5x + 4][x^2 + 5x + 6] + 1
 \end{aligned}$$

$$\text{Let } y = x^2 + 5x$$

$$\begin{aligned}
 &= (y + 4)(y + 6) + 1 \\
 &= y(y + 6) + 4(y + 6) + 1 \\
 &= y^2 + 6y + 4y + 24 + 1 \\
 &= y^2 + 10y + 25 \\
 &= y^2 + 5y + 5y + 25 \\
 &= y(y + 5) + 5(y + 5) \\
 &= (y + 5)(y + 5) \\
 &= (y + 5)^2 \\
 &= (x^2 + 5x + 5)^2
 \end{aligned}$$

(ii) $(x + 2)(x - 7)(x - 4)(x - 1) + 17$

$$(x + 2)(x - 7)(x - 4)(x - 1) + 17$$

$$2 - 7 = -5 = -4 - 1$$

$$\begin{aligned}
 &= [(x + 2)(x - 7)][(x - 4)(x - 1)] + 17 \\
 &= [x(x - 7) + 2(x - 7)][x(x - 1) - 4(x - 1)] + 17 \\
 &= [x^2 - 7x + 2x - 14][x^2 - x - 4x + 4] + 17 \\
 &= [x^2 - 5x - 14][x^2 - 5x + 4] + 1
 \end{aligned}$$

$$\text{Let } y = x^2 - 5x$$

$$\begin{aligned}
 &= (y - 14)(y + 4) + 17 \\
 &= y(y + 4) - 14(y + 4) + 17 \\
 &= y^2 + 4y - 14y - 56 + 17 \\
 &= y^2 - 10y - 39 \\
 &= y^2 - 13y + 3y - 39 \\
 &= y(y - 13) + 3(y - 13) \\
 &= (y - 13)(y + 3) \\
 &= (x^2 - 5x - 13)(x^2 - 5x + 3)
 \end{aligned}$$

(iii) $(2x^2 + 7x + 3)(2x^2 + 7x + 5) + 1$

$$(2x^2 + 7x + 3)(2x^2 + 7x + 5) + 1$$

$$\text{Let } y = 2x^2 + 7x$$

$$\begin{aligned}
&= (y+3)(y+5) + 1 \\
&= y(y+5) + 3(y+5) + 1 \\
&= y^2 + 5y + 3y + 15 + 1 \\
&= y^2 + 8y + 16 \\
&= y^2 + 4y + 4y + 16 \\
&= y(y+4) + 4(y+4) \\
&= (y+4)(y+4) \\
&= (y+4)^2 \\
&= (2x^2 + 7x + 4)^2
\end{aligned}$$

$$\begin{aligned}
\text{(iv)} \quad &(3x^2 + 5x + 3)(3x^2 + 7x + 5) - 3 \\
&(3x^2 + 5x + 3)(3x^2 + 5x + 5) - 3
\end{aligned}$$

$$\begin{aligned}
\text{Let } y &= 3x^2 + 5x \\
&= (y+3)(y+5) - 3 \\
&= y(y+5) + 3(y+5) - 3 \\
&= y^2 + 5y + 3y + 15 - 3 \\
&= y^2 + 8y + 12 \\
&= y^2 + 6y + 2y + 12 \\
&= y(y+6) + 2(y+6) \\
&= (y+6)(y+2) \\
&= (3x^2 + 5x + 6)(3x^2 + 5x + 2)
\end{aligned}$$

$$\begin{aligned}
\text{(v)} \quad &(x+1)(x+2)(x+3)(x+6) - 3x^2 \\
&(x+1)(x+2)(x+3)(x+6) - 3x^2 \\
&1 \times 6 = 6 = 2 \times 3
\end{aligned}$$

$$\begin{aligned}
&= [(x+1)(x+6)][(x+2)(x+3)] - 3x^2 \\
&= [x(x+6) + 1(x+6)][x(x+3) + 2(x+3)] - 3x^2 \\
&= [x^2 + 6x + x + 6][x^2 + 3x + 2x + 6] - 3x^2 \\
&= [x^2 + 7x + 6][x^2 + 5x + 6] - 3x^2 \\
&= [x^2 + 6 + 7x][x^2 + 6 + 5x] - 3x^2
\end{aligned}$$

$$\begin{aligned}
\text{Let } y &= x^2 + 6 \\
&= (y+7x)(y+5x) - 3x^2 \\
&= y(y+5x) + 7x(y+5x) - 3x^2 \\
&= y^2 + 5xy + 7xy + 35x^2 - 3x^2 \\
&= y^2 + 12xy + 32x^2 \\
&= y^2 + 8xy + 4xy + 32x^2 \\
&= y(y+8x) + 4x(y+8x) \\
&= (y+8x)(y+4x) \\
&= (x^2 + 6 + 8x)(x^2 + 6 + 4x) \\
&= (x^2 + 8x + 6)(x^2 + 4x + 6)
\end{aligned}$$

$$\begin{aligned}
\text{(vi)} \quad &(x+1)(x-1)(x+2)(x-2) - 16x^2 \\
&(x+1)(x-1)(x+2)(x-2) - 16x^2 \\
&1 \times 2 = 2 = -1 \times -2 \\
&= [(x+1)(x+2)][(x-1)(x-2)] - 16x^2 \\
&= [x(x+2) + 1(x+2)][x(x-2) - 1(x-2)] - 16x^2 \\
&= [x^2 + 2x + x + 2][x^2 - 2x - x + 2] - 16x^2 \\
&= [x^2 + 3x + 2][x^2 - 3x + 2] - 16x^2 \\
&= [x^2 + 2 + 3x][x^2 + 2 - 3x] - 16x^2
\end{aligned}$$

$$\begin{aligned}
\text{Let } y &= x^2 + 2 \\
&= (y+3x)(y-3x) - 16x^2
\end{aligned}$$

$$\begin{aligned}
&= y(y-3x) + 3x(y-3x) - 16x^2 \\
&= y^2 - 3xy + 3xy - 9x^2 - 16x^2 \\
&= y^2 - 25x^2 \\
&= (y)^2 - (5x)^2 \\
&= (y+5x)(y-5x) \\
&= (x^2 + 2 + 5x)(x^2 + 2 - 5x) \\
&= (x^2 + 5x + 2)(x^2 - 5x + 2)
\end{aligned}$$

3. Factorize:

$$\text{(i)} \quad 8x^3 + 12x^2 + 6x + 1$$

$$\begin{aligned}
&8x^3 + 12x^2 + 6x + 1 \\
&= (2x)^3 + 3(2x)^2(1) + 3(2x)(1)^2 + (1)^3 \\
&= (2x+1)^3
\end{aligned}$$

$$\text{(ii)} \quad 27a^3 + 108a^2b + 144ab^2 + 64b^3$$

$$\begin{aligned}
&27a^3 + 108a^2b + 144ab^2 + 64b^3 \\
&= (3a)^3 + 3(3a)^2(4b) + 3(3a)(4b)^2 + (4b)^3 \\
&= (3a+4b)^3
\end{aligned}$$

$$\text{(iii)} \quad x^3 + 18x^2y + 108xy^2 + 216y^3$$

$$\begin{aligned}
&x^3 + 18x^2y + 108xy^2 + 216y^3 \\
&= (x)^3 + 3(x)^2(6y) + 3(x)(6y)^2 + (6y)^3 \\
&= (x+6y)^3
\end{aligned}$$

$$\text{(iv)} \quad 8x^3 - 125y^3 + 150xy^2 - 60x^2y$$

$$\begin{aligned}
&8x^3 - 125y^3 + 150xy^2 - 60x^2y \\
&= 8x^3 - 60x^2y + 150xy^2 - 125y^3 \\
&= (2x)^3 - 3(2x)^2(5y) + 3(2x)(5y)^2 + (5y)^3 \\
&= (2x-5y)^3
\end{aligned}$$

3. Factorize:

$$\text{(i)} \quad 125a^3 - 1$$

$$\begin{aligned}
&125a^3 - 1 \\
&= (5a)^3 - (1)^3 \\
&= (5a-1)[(5a)^2 + (5a)(1) + (1)^2] \\
&= (5a-1)(25a^2 + 5a + 1)
\end{aligned}$$

$$\text{(ii)} \quad 64x^3 + 125$$

$$\begin{aligned}
&64x^3 + 125 \\
&= (4x)^3 + (5)^3 \\
&= (4x+5)[(4x)^2 + (4x)(5) + (5)^2] \\
&= (4x+5)(16x^2 + 20x + 25)
\end{aligned}$$

$$\text{(iii)} \quad x^6 - 27$$

$$\begin{aligned}
&x^6 - 27 \\
&= (x^2)^3 - (3)^3 \\
&= (x^2-3)[(x^2)^2 + (x^2)(3) + (3)^2] \\
&= (x^2-3)(x^4 + 3x^2 + 9)
\end{aligned}$$

$$\text{(iv)} \quad 1000a^3 + 1$$

$$\begin{aligned}
&1000a^3 + 1 \\
&= (10a)^3 + (1)^3 \\
&= (10a+1)[(10a)^2 + (10a)(1) + (1)^2]
\end{aligned}$$

$$= (10a + 1)(100a^2 + 10a + 1)$$

$$(v) 343x^3 + 216$$

$$\begin{aligned} & 343x^3 + 216 \\ &= (7x)^3 + (6)^3 \\ &= (7x + 6)[(7x)^2 - (7x)(6) + (6)^2] \\ &= (7x + 6)(49x^2 - 42x + 36) \end{aligned}$$

$$(v) 27 - 512y^3$$

$$\begin{aligned} & 27 - 512y^3 \\ &= (3)^3 - (8y)^3 \\ &= (3 - 8y)[(3)^2 + (3)(8y) + (8y)^2] \\ &= (3 - 8y)(9 + 24y + 64y^2) \end{aligned}$$

Muhammad Tayyab (GHS Christian Daska)